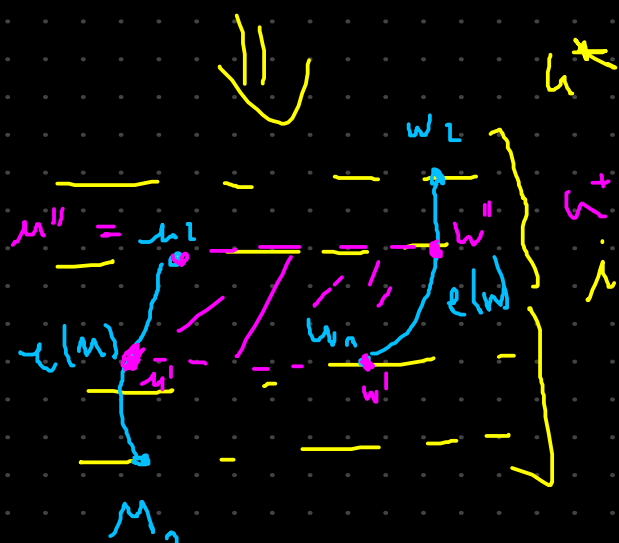


$$\rho \text{ of } S(h) = \begin{cases} vw = uv \\ vw = \neg wv \end{cases}$$

$$m, w \in h, \quad l(m) = l(w)$$

$$\xrightarrow{m} \xrightarrow{w} \xrightarrow{h} i$$



$e(m), e(w)$ are critical by construction

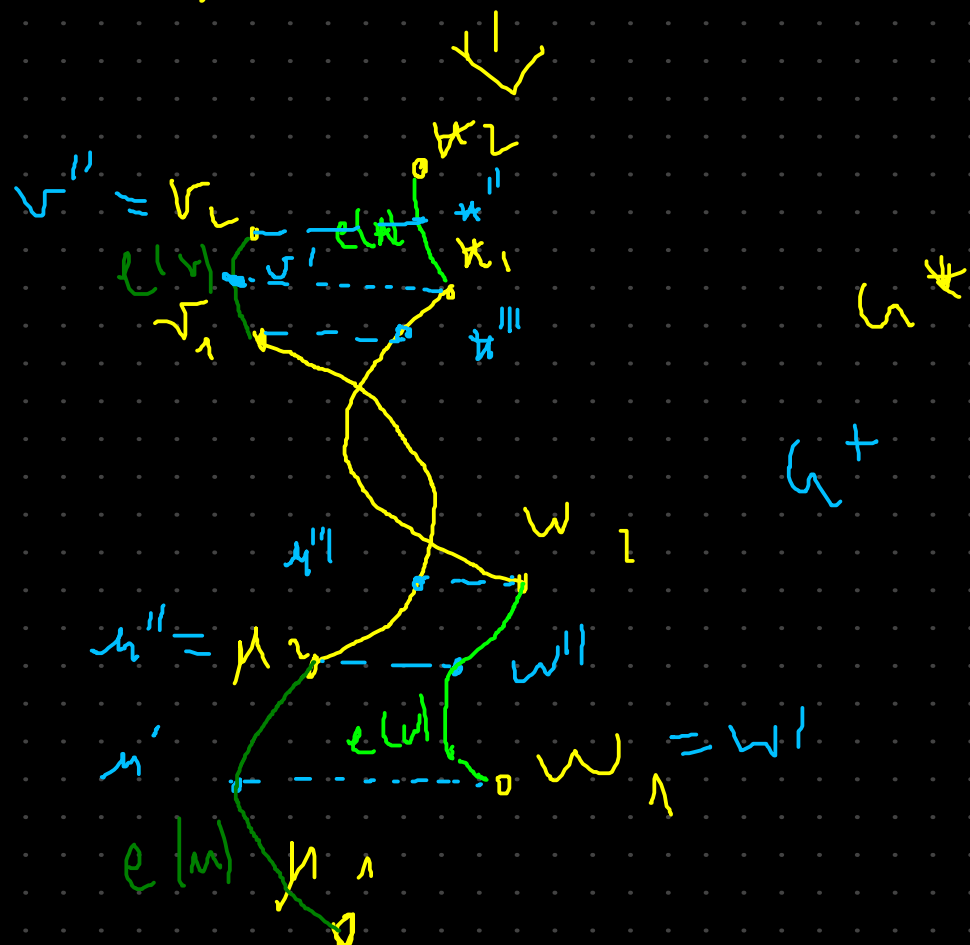
$$m_1 < w_1, \quad w_1 < m_2$$

$$\text{Set } \varphi(mw) = \varphi^+(m'w')$$

$$\varphi^+(m'w') = \varphi^+(m''w'')$$

$$(m, v), (w, x) \in E(h) \quad / \quad l(w) = l(w)$$

$(m', m''), (w', w'')$ be the limits of $e(m), e(w)$ in h^+ .



Lemma 4

h p.l.o, φ of $S(h)$

$\Rightarrow$ HT drawing Γ^* of h^*

$\Rightarrow$ level plane Γ^+ of $h^+ \equiv h$

(proof)

φ^+ of $S(h^+)$ $\longleftarrow$ φ of $S(h)$

let $u \in h^*$ and $l(u) = i$

$\underbrace{\quad u \quad \quad w \quad \quad}_{\varphi^+(uw) = \text{true}}$

let Γ^* of h^* from h^+



$v, w, u, x \in h^*$

L^2

$\varphi^+(w'w')$
 $= \varphi([v'x'])$

on their level.

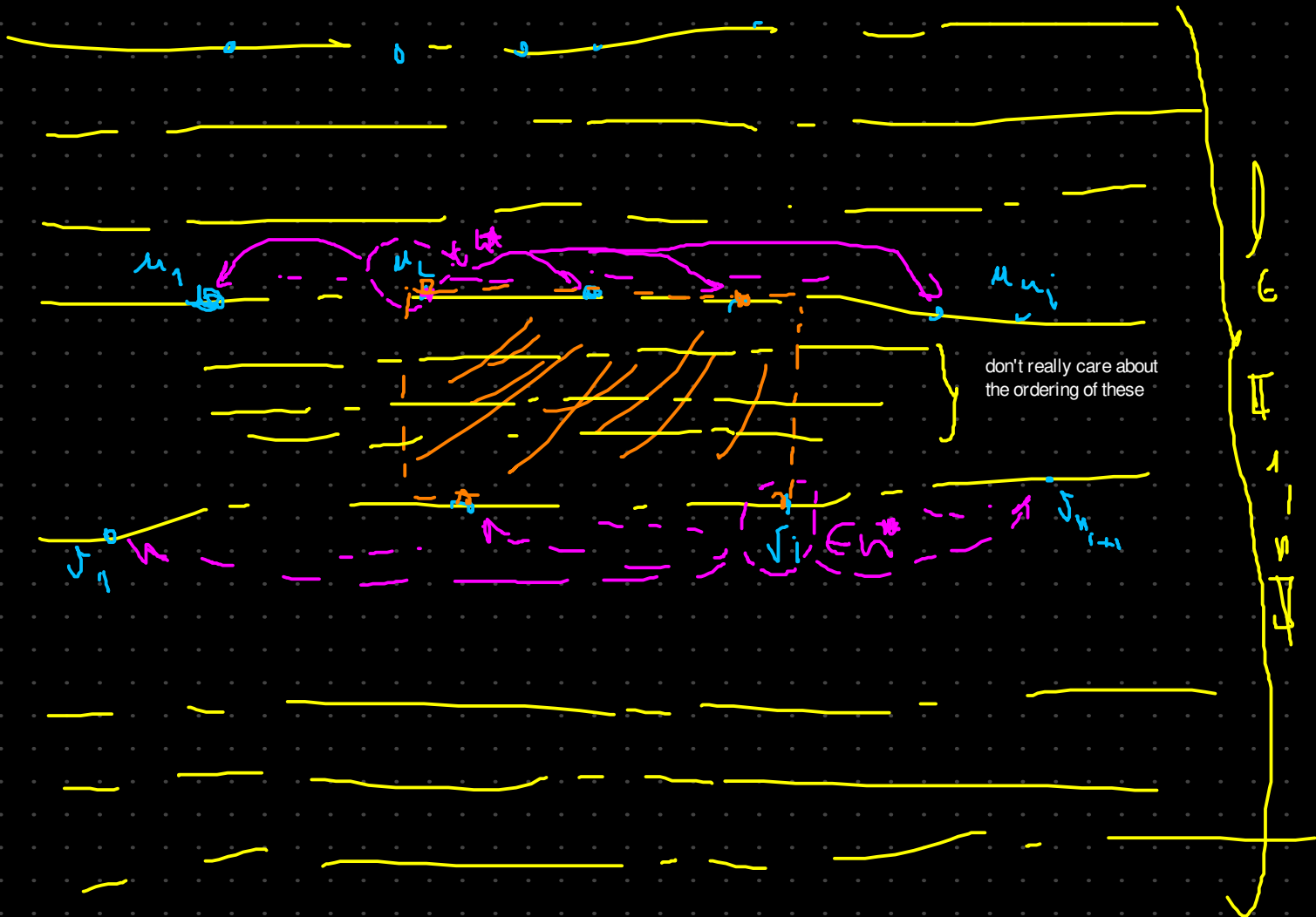
$$\varphi(u, v) = \varphi(v, u) \quad \forall (u, v), (w, x) \in E(h) \text{ in } h^+ \\ / u, v, w, x \in V(h)$$

$$\varphi(u, v) = \varphi(v, u) \quad \forall u, v \in V(h)$$

h is proper level graph

let u'', w'' two vertices of h^+
with $\ell(u'') = \ell(w'')$

$$\varphi = \zeta(h)$$



$$\forall u, w \in h^+ \quad \varphi(u, w) = \varphi^+(u, w)$$

$m, w \in L \quad / \quad l(m) = l(w)$
 $e(w), e(m)$ are oriented in L^*

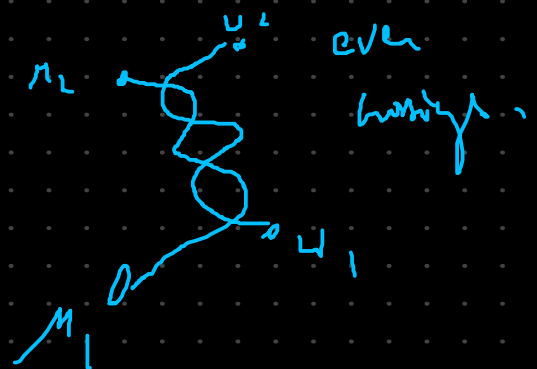
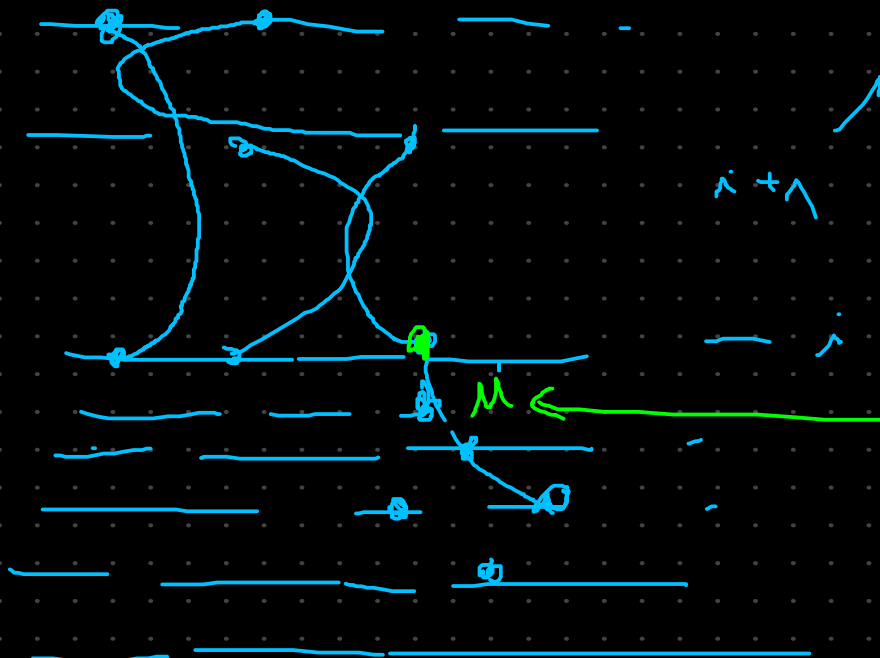


$$\varphi(mw) = \varphi + l(m'w')$$

$m_1, m_2, w_1, w_2 \in L^*$

and it's a HT dr.

so if we have a vertex of G^* then we are sure that the vertex doesn't exist in between even crossings so it's relative order should be the same as the truth assignment of $S(G)$

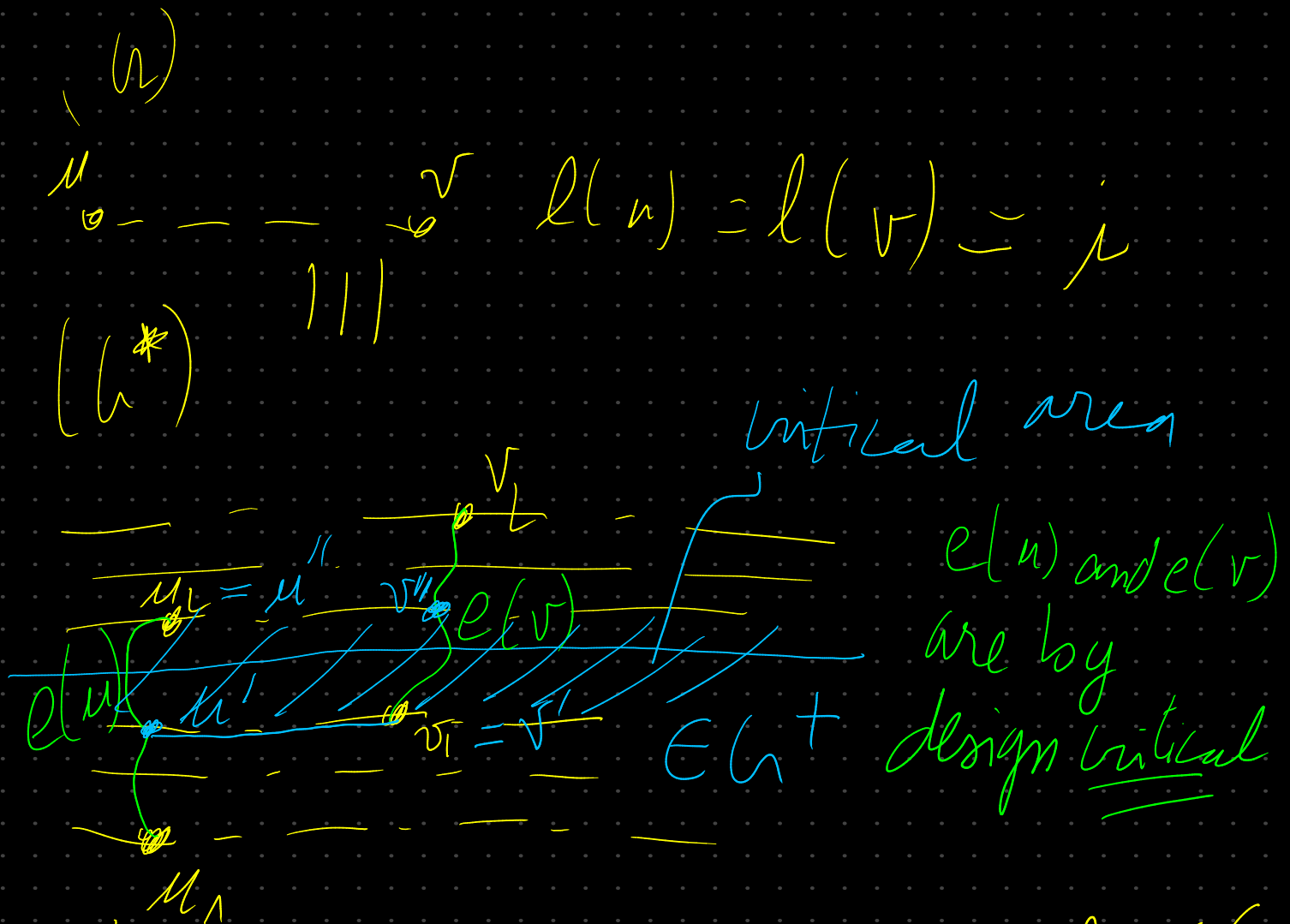


if u is considered part of G^* then, the only incoming edge is the stretch edge, and the other endpoint of the edge doesn't have any other neighbors.

⇒ Lemma 3 part used in Lemma 4 —

* good.

— construct φ of $S(h)$ as follows:



φ^+ is a satisfying assignment for $S(h^+)$

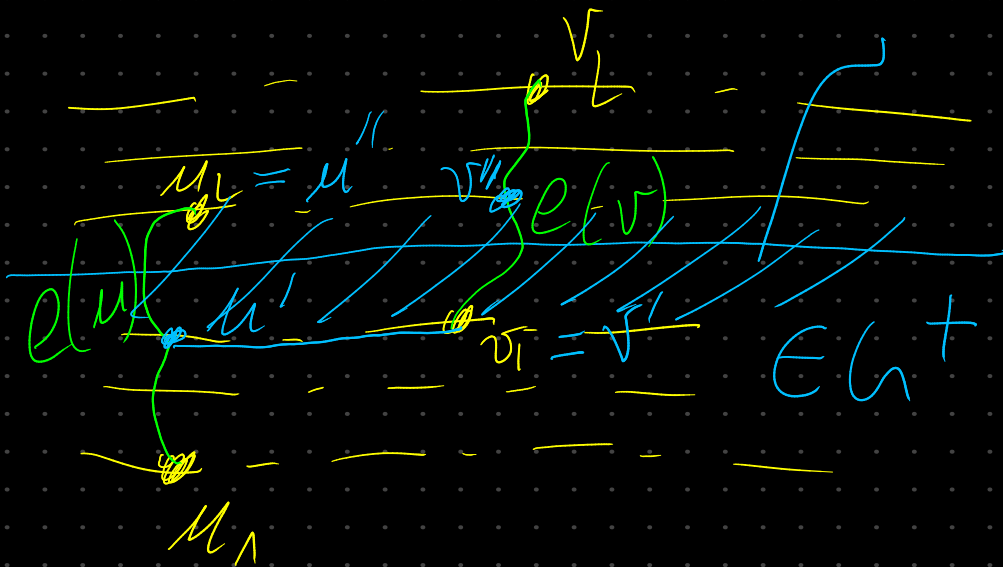
⇒ Crossing clauses are satisfied
 $uw \Leftrightarrow vx \quad (u, v), (w, x) \in E(h)$

and $\ell(u) = \ell(w)$
 $\ell(v) = \ell(x) \quad / x \in V(u)$

so if $(u, v), (w, v) \in E(G)$
and are independent then

$$\varphi^+(u, v) = \varphi^+(v, u)$$

if we go back to our example:



$$\varphi^+(m'', v'') = \varphi^+(m', v')$$

$$\varphi^+(u'', v'') = \varphi^+(u', v')$$

and the same is true for all the subdivisions of $e(u)$ and $e(v)$

$$\forall y \in e(u), z \in e(v)$$

$$\varphi^+(u'', v'') = p^+(y, z)$$

φ^+ also satisfies the $\forall y (\Rightarrow \exists y \neg$
 clause then (consistency)

$$\varphi^+(u, v) = \neg \varphi^+(v, u)$$

$$\varphi''(u, v) \quad \varphi''(v, u)$$

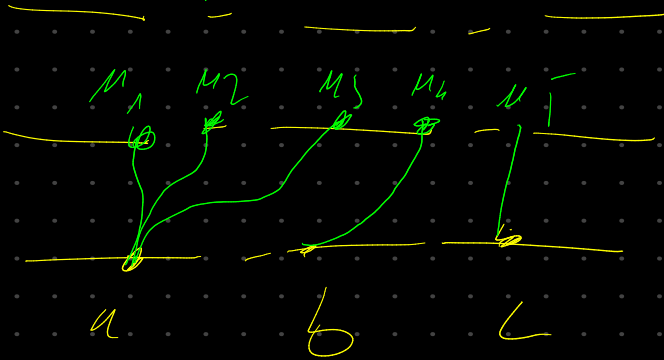
$$\Rightarrow \varphi(u, v) = \neg \varphi(v, u)$$

so φ also satisfies the consistency clause
 of $S(h)$.

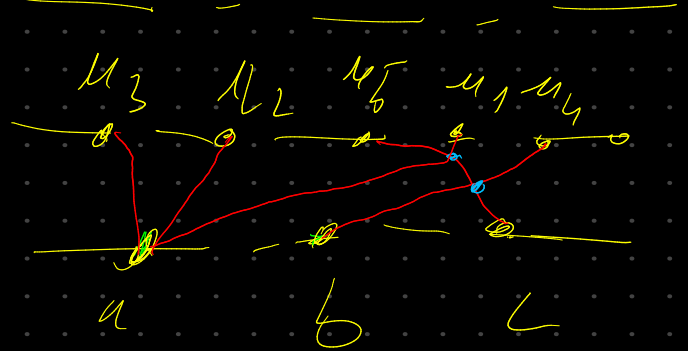
example: $\{a, b, c\} \in V(h)$

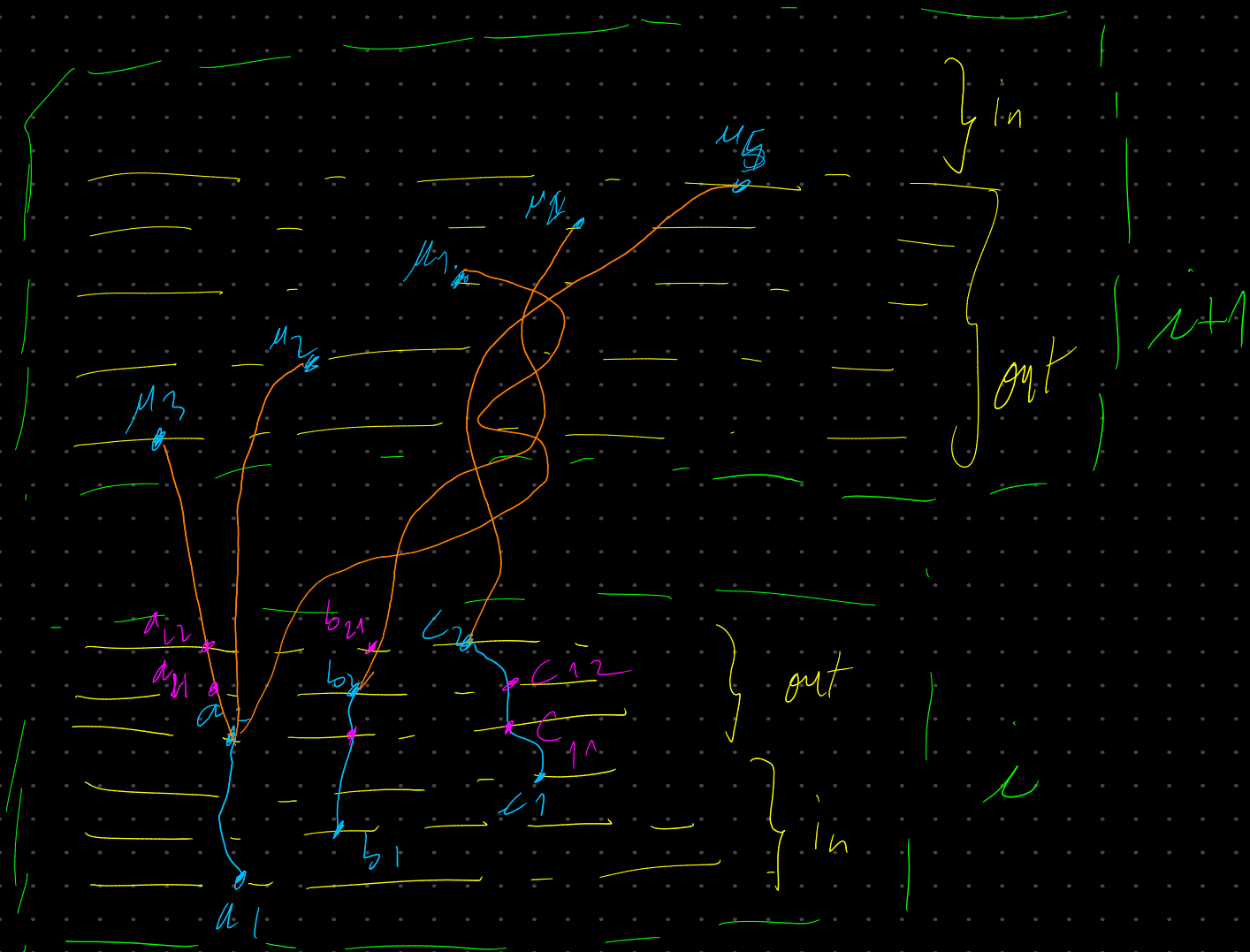
with $\ell(a) = \ell(b) = \ell(c)$

$\Gamma_{opt}(h)$



$\Gamma(h)$





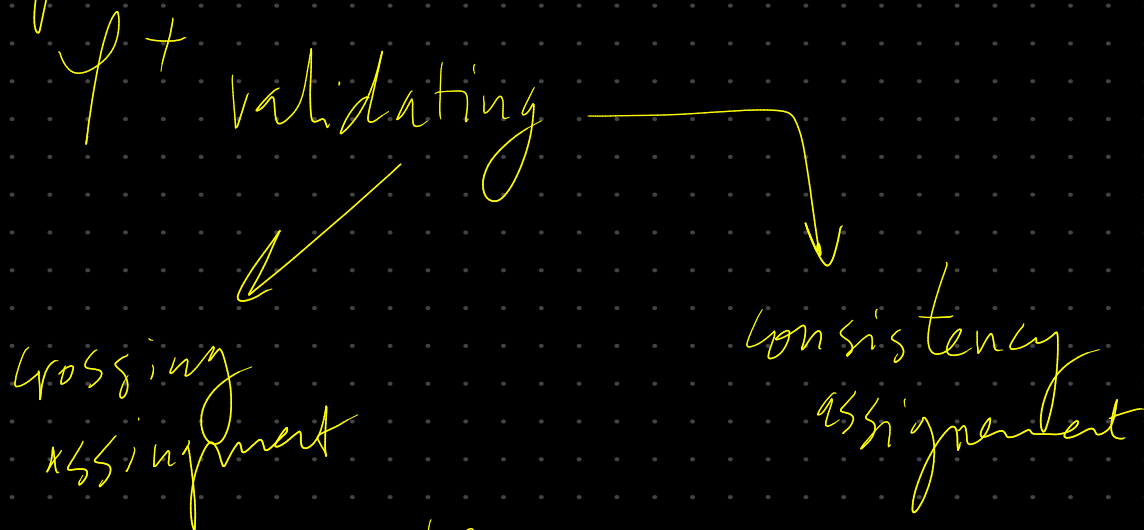
$$\varphi(ab) = \varphi^+ \begin{matrix} a_1 & a_2 & a_{12} \\ b_1 & b_2 & b_{12} \end{matrix}$$

$$\varphi(bL) = \varphi^+ \begin{matrix} b_1 & b_2 & b_{12} \\ c_1 & c_2 & c_{12} \end{matrix}$$

$$\varphi(aL) = \varphi^+ \begin{matrix} a_1 & a_2 & a_{12} \\ c_1 & c_2 & c_{12} \end{matrix}$$

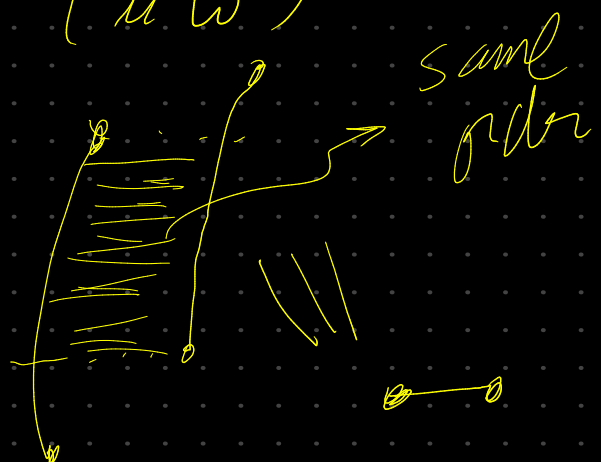
I think, I got a better grasp of the idea in this part of the proof:

we have φ^+ truth assignment of $S(h^+)$ and we want to map it to our original h , meaning mapping $V(h^+) \rightarrow V(h)$ and also prove that we can map to a truth assignment of $S(h)$.



the mapping

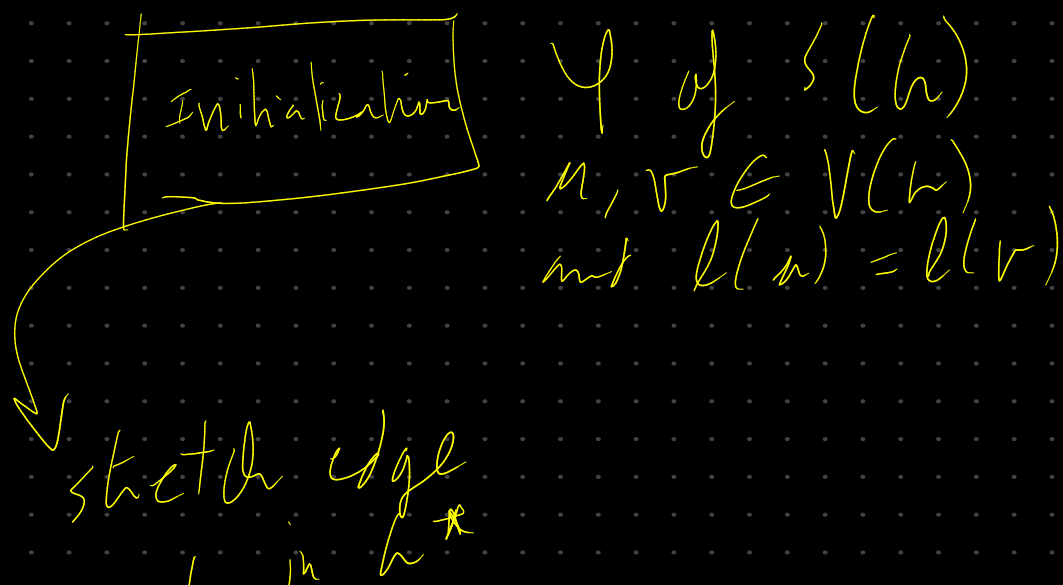
$$\varphi(uv) = \varphi^+(u'v')$$



so what do I need for lemma 4

- I have φ , I need to map it to φ^+ and also prove that φ^+ is also a truth assign. but for $S(h^+)$

Proof of Lemma 3 scheme



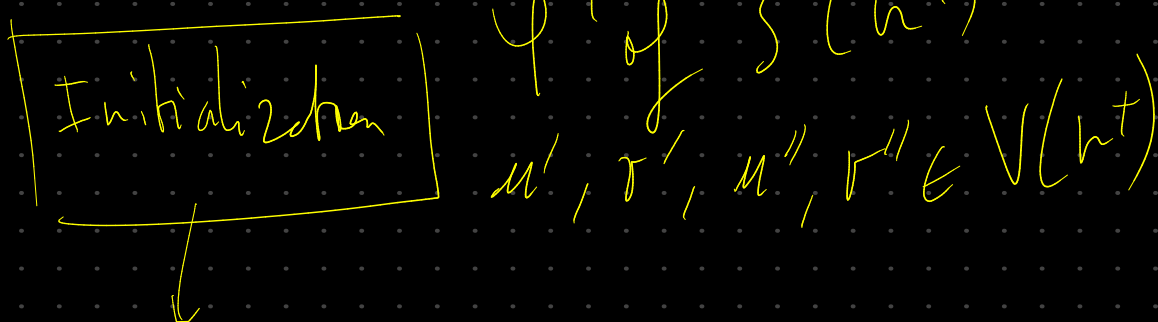
limits of $h^+ \rightarrow$ Mapping

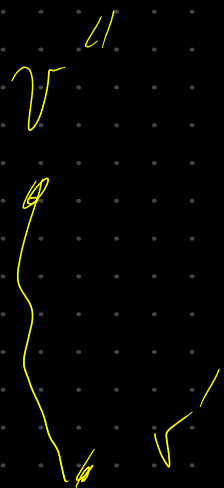
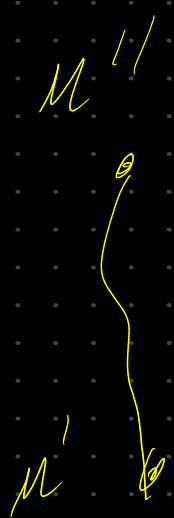
(concl.)

no crossing

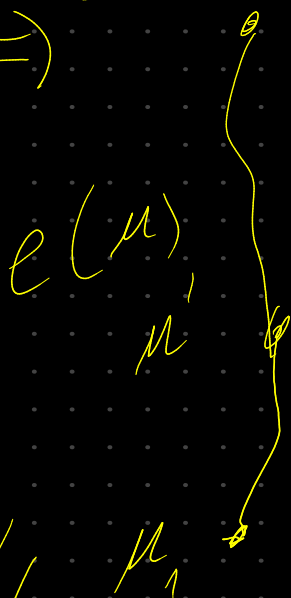
consistency

so for lemma 4:





$$\mu_2 = \mu'' \Rightarrow$$



$e(u)$ and $e(w)$ are critical
 $\hookrightarrow \varphi(u, w) = \varphi^+(u', w')$
 $\varphi(u, w)$

no
 universe

$$\varphi^+(u, w) = \varphi(u, w)$$

$$\varphi^+(x, x) = \varphi(v, x)$$

Consistency

$$\begin{cases} \varphi^+(u, w) = \varphi(u, w) \\ \varphi^+(w, u) = \varphi(w, u) \\ \varphi(u, w) = \neg \varphi(w, u) \\ \varphi^+(u, w) = \neg \varphi^+(w, u) \end{cases}$$

So we can construct truth assign. φ^+ of $S(h^+)$ from the truth assign. φ of $S(h)$

* next step: φ^+ in h^+ to a HT Γ^*

We must prove that we can end up with a h^* HT-drawing $\rightarrow$ one vertex per level
 $\rightarrow$ even crossings.

let $u \in h^*$ in level i

$\frac{w_2}{\circ} - \text{---} u - \frac{w_1}{\circ}$

Γ^* drawing of h^* that includes Γ^+

$v' = v$ x'
 $u' = u$ $w = w'$ (h^+) we know that (each of) the 2 pairs (u', w') and (v', x') have exactly one vertex in h^* (that's how the ribbon zone is constructed)

From Lemma(2) and because φ^+ is a t.A of $S(h^+)$, meaning that $\varphi^+(u'w') = \varphi(v'x')$ implies that

(u, v) and (w, x) intersect an even number of times in Γ^*

and this is true for critical edges.

$\Rightarrow \exists \text{ HT. } \Gamma^* \text{ of } h^*$